

Graph-Structured Memory for Cognitive Agents: From Knowledge Graphs to Non-Euclidean Geometry

Michael Banf, Johannes Kuhn, Slawomir Garcarz, Liliya Imasheva, Dominik Filipiak, Marc Stiller, Christian Mader, Gustaw Kalek, Maximilian Schattauer, Anton Steuer, Gzregorz Luczyna, Konstantin Gärtner, Emily Rasenecker, Max Becker and Sebastian Fetz

Perelyn GmbH, Munich, Germany

Abstract

Large language model based agents require persistent memory that spans sessions, tracks temporal evolution, and supports multi-hop reasoning. As context windows grow, the naïve strategy is to compress all prior interaction into the prompt, however compression discards relational structure. Graph-structured memory offers a fundamentally different approach: store knowledge in a graph and retrieve selectively. Yet current systems embed these graphs exclusively in Euclidean space, incurring distortion on the hierarchical, tree-like structures inherent to agent memory. We survey research papers between 2023 and 2026, identifying six architectural patterns and a retrieval spectrum from training-free graph algorithms to graph neural network message passing. We then analyze how non-Euclidean geometric techniques, such as discrete Ricci curvature for diagnosing retrieval bottlenecks, hyperbolic embeddings for hierarchy-preserving retrieval, Lie group manifolds for temporal knowledge graph fusion, and mixed-curvature product spaces, are already proven in retrieval and KG reasoning, with significant gains over Euclidean baselines. We argue that since agent memory is fundamentally retrieval over a persistent knowledge graph, these techniques are directly applicable as foundational layers, but integrating them into cross-session memory architectures is the defining open challenge.

Keywords

Knowledge Graphs, Agent Memory, Non-Euclidean Geometry, Hyperbolic Embeddings, Temporal Knowledge Graphs, LLM Agents

1. Introduction

The dominant memory pattern for Large Language Model (LLM) agents is stateless: each session begins from scratch. When persistence is added, it typically takes one of two forms: i) flat extraction, i.e. triplet stores, key-value memories, or embedding-indexed chunks [2, 3]; or ii) brute-force compression of all prior interaction into the context window, which grows ever larger but still cannot preserve relational structure. Both handle factual recall but fail systematically on temporal reasoning, multi-hop inference, and knowledge evolution. Flat paradigms—vector stores, key-value extraction, and even the emerging LLM Wiki pattern [53]—lack native support for multi-hop traversal, temporal ordering, causal structure, and hierarchical consolidation; graph-based architectures provide all of these by construction. Benchmark evidence quantifies the deficit: LoCoMo-Plus [4] reports cognitive accuracy below 25% across Mem0, Letta, and Zep, with a 35–45 pp drop from factual to cognitive tasks. MemoryAgentBench [5] finds all methods below 7% on multi-hop conflict resolution, while multi-strategy systems combining graph traversal with vector search achieve 91.4% retrieval accuracy versus 49.0% for extraction-based systems [6]. Graph-structured memory offers a fundamentally different strategy: instead of compressing everything into the context window, store knowledge in a graph and retrieve selectively. Knowledge graphs (KG) ground entities and relations while temporal KGs add validity intervals; hierarchical event graphs capture episodic consolidation. The core challenge then shifts from *what to discard* to *how to find the right subgraph*, i.e. a retrieval problem where graph topology is a first-class signal. Here we survey the current state of the art graph-based agent memory systems between 2023–2026 [7, 8], with three significant findings standing out: i) entity-relation KGs account for 60% of surveyed graph-memory

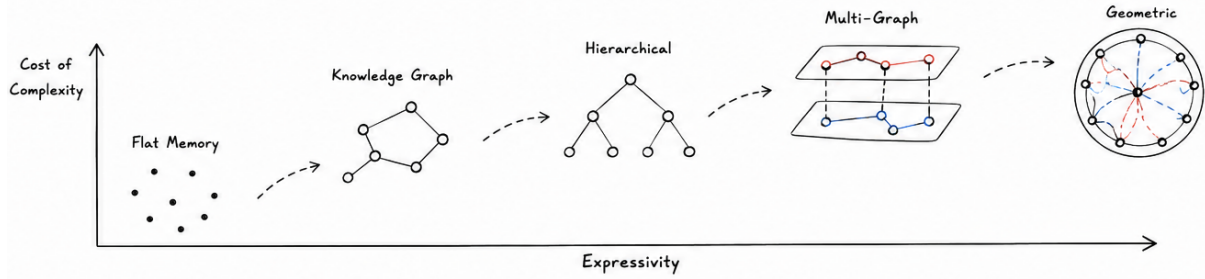


Figure 1: Structural progression of agent memory representations. Each level adds expressivity at the cost of complexity. Current production systems operate at the knowledge graph and hierarchical levels, while geometric representations remain a largely unexplored frontier.

papers; non-KG structures remain rare; ii) 57% implement no temporal mechanism, constituting a field-wide deficit; and iii) retrieval is overwhelmingly training-free, i.e. graph algorithms and multi-strategy heuristics power every production system, while only a tiny fraction of surveyed research work applies graph neural networks (GNN) message passing.

Figure 1 illustrates the structural progression from flat memory to geometric representations. This paper contributes: (1) a taxonomy of graph-based agent memory patterns including temporal grounding; (2) a retrieval spectrum with a model-size equalizer analysis; (3) an analysis of non-Euclidean geometric techniques and their application to agent memory challenges; and (4) a research agenda.

2. Taxonomy of Graph-Based Agent Memory

From our classification we identify four established architectural patterns, two emerging ones, and a cross-cutting temporal dimension (Table 1). The most prevalent pattern is the *entity-relation knowledge graph*, representing memory as typed (*subject, relation, object*) triples. CraniMem [13] combines a bounded FIFO episodic buffer with an entity-relation KG. UltRAG [14] queries large-scale KGs via a pre-trained GNN foundation model. AriGraph [15] integrates episodic and semantic memory within a single KG. GraphRAG [12] clusters via Leiden into hierarchical community summaries. Entity-relation KGs excel at factual memory but are limited to binary relations.

The *Hierarchical event graphs* pattern addresses this by maintaining episodic and semantic layers with cross-layer edges. GAM [16] combines a Topic Associative Network with Event Progression Graphs. MemFly [17] implements a three-layer hierarchy via the Information Bottleneck principle. [18] introduces Cue-Tag-Content heterogeneous graphs. MemGraphRAG [1] organizes memory into three layers (ontology schemas, entity-relation facts, source passages) with multi-agent self-adjudication, i.e. a conflict detection agent identifies inconsistencies and a resolution agent adjudicates via provenance evidence, demonstrating that filtering low-frequency triples actually improves accuracy on multi-hop Question Answering.

Bipartite and associative graphs take a lighter approach. For example, LP-RAG [19] constructs a bipartite chunk-query graph with mutual k -NN edges (retrieval via GNN link prediction outperforms embedding similarity by 1.6–2.8% EM), while AssoMem [20] anchors dialogue to automatically extracted clues, i.e. minimal overhead, but limited representational power.

At the other extreme, *multi-graph composites* maintain several orthogonal views simultaneously. MAGMA [21] connects event-nodes via four edge types (temporal, causal, semantic, entity) with dual-stream fast/slow consolidation. TM2RAG [22] extends this with five graphs unified through a meta graph.

Two emerging patterns push beyond pairwise relations. *Hypergraph memory* systems, such as HyperGraphRAG [23], HyperMem [24], HGMem [25], use hyperedges for n -ary facts and multi-step reasoning chains, though all treat the hypergraph as a data structure without higher-order message passing. *Procedural strategy graphs* encode agent trajectories directly: Trainable Graph Memory (TGM) [26] represents

Table 1

Comparison of graph-based agent memory systems. Temporal: ✓ = deep mechanism, △ = timestamps/decay, — = none. Retrieval: GA = graph algorithms, MS = multi-strategy, RL = reinforcement learning, GNN = message passing, AB = agent-based.

System	Graph Type	Temp.	Retr.	Evolution	Best Score	Benchmark
<i>Production & multi-strategy systems</i>						
Zep/Graphiti	KG (bi-temporal)	✓	MS	Temporal invalidation	94.8%	Deep Mem. Retrieval
Hindsight	KG + vector	△	MS	—	91.4%	LongMemEval
FalkorDB+Mem0	KG (graph DB)	—	MS	Dedup/merge	<140 ms p99	Production
<i>Hierarchical & multi-graph architectures</i>						
GAM	Hierarchical (2-layer)	△	GA	State-based consol.	40.00 F1	LoCoMo (7B)
MAGMA	Multi-graph (4 types)	✓	GA	Dual-stream consol.	0.700	LoCoMo Judge
MemFly	Hierarchical (3-layer)	—	GA	IB clustering	44.4 F1	LoCoMo (GPT-4o)
MemGraphRAG	Hier. (3-layer)	—	MS	Schema filter + adjud.	59.25%	Multi-hop QA
<i>Entity-relation KG systems</i>						
Mem0	KG (entity-relation)	—	GA	Dedup/merge	28.86 F1	LoCoMo (7B)
HippoRAG	KG (open)	—	GA	—	+3–10 F1	Multi-hop QA
CraniMem	KG (gated buffer)	—	GA	Replay consol.	−0.011 F1	HotpotQA noise
<i>Temporal-aware systems</i>						
RecallM	KG (temporal window)	✓	GA	Context revision	4× vs Vec	Temporal QA
MemoTime	KG (Tree of Time)	✓	GA	Temporal monoton.	—	Temporal reason.
TReMu	KG (neuro-symbolic)	✓	GA	Code-gen dates	—	Multi-session
<i>RL & learned retrieval systems</i>						
TGM	Procedural (3-layer)	—	RL	REINFORCE weights	0.426 EM	HotpotQA (4B)
Memory-R1	KG	—	RL	RL operations	—	Memory via RL
<i>GNN & higher-order systems</i>						
LP-RAG	Bipartite	—	GNN	—	+2.8% EM	Multi-hop QA
MemoGraph	KG + RGCN	—	GNN	—	86.9%	MATH (7B)
HyperMem	Hypergraph (3-level)	—	GA	—	—	—
<i>Multi-agent & procedural systems</i>						
Collabor Mem.	KG (shared)	—	AB	Access control	—	Multi-agent
PRAXIS	State-action tuples	—	GA	Real-time learn.	44.1%	REAL (web)
SkillRL	Hier. SkillBank	—	RL	Recursive evol.	+14%	ALFWorld
SimpleMem	Multi-view index	—	MS	Sem. compression	+26.4% F1	LoCoMo

trajectories as a heterogeneous DAG optimized via REINFORCE, where the trained Qwen3-4B outperforms baseline Qwen3-8B, i.e. procedural graph memory substituting for model scale. PRAXIS [27] implements procedural memory for web agents. SkillRL [28] builds a hierarchical SkillBank.

Cutting across all patterns is the question of *temporal grounding*. Despite being the central promise of graph-based memory, 57% of surveyed papers implement no temporal mechanism and only 20% go beyond timestamps. Graphiti/Zep [10] provides the most complete solution via a bi-temporal model distinguishing event time from ingestion time, where every edge carries validity intervals. RecallM [29] uses temporal windows with Hebbian-inspired relation strength for a more effective belief updating than vector databases. MemoTime [30] organizes temporal reasoning into time trees, while TReMu [31], similarly to Graphiti/Zep, decouples mention time from event time. Biologically inspired mechanisms offer implicit temporal grounding. For example, Mnemosyne [32] applies exponential decay with spreading activation. MIRROR [54] validates fast encoding plus slow consolidation. MARTA [55] uses predictive entropy to arbitrate retrieval. However, a staleness gap persists, i.e. no system tracks when abstracted semantic knowledge becomes outdated.

3. From Graph Algorithms to Learned Retrieval

Memory retrieval over graphs spans a spectrum from training-free algorithms to fully learned neural approaches. The dominant approaches require no training. For example HippoRAG [9] implements Personalized PageRank as a neurobiologically inspired pattern-completion mechanism. GraphRAG [12] and MemFly use Leiden for community detection. GAM applies exponential recency decay. These

algorithms work on any graph at 10–150 ms, explaining why every deployed system adopts them. Production systems go further by combining multiple pathways. E.g. Hindsight [11] executes semantic search, BM25, graph traversal, and temporal filtering simultaneously, fusing via cross-encoder. SimpleMem [33] achieves improvement over Mem0 while reducing tokens via three-stage semantic compression. Beyond heuristics, RL-based approaches learn to optimize different aspects of graph memory, such as edge weights [26], memory operations [34], graph construction [35], and unified retrieval-reasoning [36]. D-RAG [37] enables joint retriever-generator optimization through Gumbel-Softmax subgraph sampling. At the far end of the spectrum, only a tiny fraction of the surveyed systems apply GNN message passing, for example LP-RAG [19] recasts retrieval as inductive link prediction, MemoGraph [38] encodes evolving reasoning graphs. The scarcity reflects fundamental tensions, i.e. cold start, no retrieval labels, graph evolution outpacing retraining, and diminishing returns on factual tasks where heuristics already achieve high accuracy.

A striking pattern across our corpus is that graph-structured memory disproportionately benefits smaller models. GAM achieves +39% F1 improvement over Mem0 at 7B but only +7% at GPT-4o, and on LongDialQA this ratio reaches 17 times improvement. The trained Qwen3-4B with graph memory outperforms baseline Qwen3-8B showing that graph structure directly substitutes for model scale. GNN-RAG [39] shows a fine-tuned 7B model with graph retrieval matches GPT-4 on knowledge graph question answering tasks.

These results may be explained by four mechanisms: i) context window relief via focused subgraph retrieval; ii) explicit structure offloading reasoning that larger models derive implicitly; iii) token efficiency (GNN-RAG’s reduction); and iv) strategic compensation from meta-cognitive priors in graph structure. Finally, graph memory must evolve as agents accumulate experience, i.e. consolidation merges episodic observations into compressed semantic representations, and SkillRL [28] achieves recursive skill co-evolution with significant token compression versus raw trajectory storage.

4. Geometric Representations for Agent Memory

All graph-based memory systems surveyed above embed their graphs in flat Euclidean space. Yet recent work on graph neural networks, knowledge graph embeddings, and topological deep learning [56] shows that non-Euclidean representations—hyperbolic embeddings, discrete Ricci curvature, Lie group manifolds—can dramatically improve performance on hierarchical and temporal data. Since graph-structured memory is fundamentally a retrieval problem—finding the right subgraph at the right time—these techniques are directly applicable as foundational components of memory architectures.

Selective retrieval over a knowledge graph fails when the graph topology itself creates bottlenecks. *Discrete Ricci curvature* [57, 40] provides a per-edge diagnostic for this: edges within tight clusters have positive curvature, while edges bridging distant communities have negative curvature. Negatively curved edges cause *over-squashing* [58]—information from exponentially many source nodes must squeeze through a single bridge edge, causing multi-hop retrieval to suffer exponential signal decay. This offers a structural explanation for the cognitive accuracy ceiling below 25%. Three concrete applications emerge: i) Forman-Ricci curvature provides a training-free, $O(|E|)$ diagnostic that flags edges where retrieval is likely to degrade; ii) Stochastic Discrete Ricci Flow (SDRF) [40] offers a principled geometric alternative to Leiden-based community detection for hierarchical consolidation; and iii) structural lifting from graphs to hypergraphs [59, 41] can promote bottleneck edges to hyperedges, restoring multi-hop connectivity.

Selective retrieval also fails when the embedding space itself cannot separate what the graph distinguishes. As memory grows, Euclidean spaces suffer from *semantic collapse*: hierarchical structures are compressed into a low-dimensional subspace because Euclidean volume grows polynomially while tree-like knowledge branching grows exponentially [60]. Gao and Metaxas [61] formalize this as semantic shift—as diversity increases, pooled representations drift away from every individual embedding. *Hyperbolic space*, where volume grows exponentially, provides a geometric remedy by naturally separating hierarchy levels: abstract topics embed near the origin, episodes in the middle ring, and

specific facts near the boundary. HyperbolicRAG [42] fuses Euclidean and hyperbolic rankings for hierarchy-aware retrieval. HypRAG [43] develops fully hyperbolic transformers in the Lorentz model. HELM [44] shows LLM token embeddings already exhibit substantial negative Ricci curvature, suggesting that LLM representations are naturally hyperbolic and motivating matching the memory embedding space to the model’s intrinsic geometry. HyperVC [62] applies variable-curvature hyperbolic spaces to temporal KG snapshots. However, HyperGCL [63] shows that naive contrastive learning in hyperbolic space still causes dimensional collapse, meaning collapse-aware objectives are required, such as HypRAG’s Outward Einstein Midpoint pooling operator. Since real memory graphs exhibit topological heterogeneity—some regions are tree-like, others cyclical or clique-dense—a single curvature may not suffice. GraphMoRE [64] addresses this with a mixture of Riemannian experts that assigns spherical, hyperbolic, or Euclidean geometry per node.

Cross-session memory further requires temporal reasoning: retrieving not just what was stored but when, and how knowledge evolved. In temporal KGs, entities, relations, and timestamps live in fundamentally different spaces, making direct fusion difficult. *Lie groups* such as $SO(n)$ (rotation matrices) offer a solution: their tangent space—the Lie algebra—is an ordinary vector space, so projecting via the logarithmic map lets heterogeneous factors be added, scaled, and processed by standard neural network layers. Li et al. [46] apply this using $SO(2)$ Givens rotations to unify temporal KG factor embeddings. TeAST [45] maps temporal relations onto Archimedean spiral geometry. TempHypE [65] unites Poincaré ball embeddings with Neural ODEs for continuous-time evolution. RicciKGE [66] co-evolves Ricci flow with KG embeddings, where curvature smooths bottleneck regions while updated embeddings reshape curvature. At the foundation-model scale, RiemannGFM [67] learns cross-domain transferable Riemannian graph representations, suggesting that zero-shot geometric encoding of memory traces may become feasible without per-agent supervised training.

These geometric techniques are already proven in retrieval and KG reasoning. The open challenge is integrating them into persistent, cross-session memory architectures with evolving graphs.

5. Open Challenges and Conclusion

Graph-structured memory reframes the persistence problem. Instead of compressing everything into the context window, store knowledge in a graph and retrieve selectively. Our survey shows this paradigm works with multi-strategy retrieval reaching top performances on long memory evaluation benchmarks, and graph memory equalizing even small models to near-frontier performance at significant lower cost. However, but several challenges remain. Current systems vary widely in graph types, node/edge semantics, and temporal models. Therefore, a unified ontology specifying how entities, events, temporal intervals, and higher-order groupings compose would enable interoperability and memory transplantation across architectures [47]. Graph algorithms dominate practice while GNN-based message passing remains limited to a handful of systems. Hence, whether learned structural representations can break the cognitive ceiling is an open question. Collaborative memory is largely unexplored, yet the same relational structure that enables rich retrieval also enables adversarial injection [50, 48, 49]. Memory quality deserves attention beyond downstream accuracy, i.e. diagnostic analysis [51] reveals that retrieval quality, not write-time sophistication, remains the primary bottleneck, and filtering low-frequency schema triples actually improves accuracy. Scalability is also a practical concern, with systems spanning wide latency ranges, and at lifetime scale, approximate retrieval, compression, and asynchronous processing [52] become critical. Finally, geometric retrieval techniques such as curvature diagnostics, hyperbolic embeddings, mixed-curvature manifolds, and continuous-time dynamics, are proven in retrieval and knowledge graph reasoning but await systematic integration into persistent, cross-session memory architectures. Addressing these challenges collectively would move the field from isolated retrieval components toward truly unified agent memory that is persistent, geometry-aware, and capable of the structured recall that cognitive continuity demands.

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